

5. Write android Program to:

a. Create a menu driven program android app that contains simple interest and area of triangle. Create separate fragment for each menu and perform task as their name.

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<FrameLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/fragment_container"
    android:layout_width="match_parent"
    android:layout_height="match_parent"/>
```

main_menu.xml

```
<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android">
    <item
        android:id="@+id/menu_si"
        android:title="Simple Interest"/>
    <item
        android:id="@+id/menu_area"
        android:title="Area of Triangle"/>
</menu>
```

Fragment_simple_interest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:padding="16dp"
    android:layout_width="match_parent"
    android:layout_height="match_parent">
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content">
```

```
        android:text="SimpleInterest Calculate"
        android:textSize="30sp"
        android:layout_gravity="center"
    />
<EditText
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:id="@+id/etP"
    android:hint="Enter the Principal"
    android:inputType="numberDecimal"
    android:textSize="25sp"/>
<EditText
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:id="@+id/etR"
    android:hint="Enter the Rate"
    android:inputType="numberDecimal"
    android:textSize="25sp"/>
<EditText
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:id="@+id/etT"
    android:hint="Enter the Time"
    android:inputType="numberDecimal"
    android:textSize="25sp"/>
<Button
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:id="@+id/btnCalc"
    android:text="Calculate "
```

```
        android:layout_gravity="center"
        android:textSize="25sp"/>
<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:id="@+id/tvResult"
    android:textSize="23sp"/>
</LinearLayout>
```

Fragment_triangle.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Area of Triangle Calculate"
        android:textSize="30sp"
        android:layout_gravity="center"/>
    <EditText
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:id="@+id/etBase"
        android:textSize="25sp"
        android:hint="Enter the Base"
        android:inputType="numberDecimal"/>
    <EditText
        android:layout_width="match_parent"
```

```
        android:layout_height="wrap_content"
        android:id="@+id/etHeight"
        android:textSize="25sp"
        android:hint="Enter the Height"
        android:inputType="numberDecimal"/>

<Button
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:id="@+id/btnCalc"
    android:textSize="23sp"
    android:text="Calculate "
    android:layout_gravity="center"/>
<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:id="@+id/tvResult"
    android:textSize="23sp"/>
</LinearLayout>
```

Fragment_SimpleInterest.java

```
package com.bhupendra.trianglesimpleinterest;

import android.os.Bundle;
import androidx.fragment.app.Fragment;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;

public class FragmentSimpleInterest extends Fragment {
```

```
EditText etP, etR, etT;

Button btnCalc;

TextView tvResult;

// TODO: Rename parameter arguments, choose names that match
// the fragment initialization parameters, e.g. ARG_ITEM_NUMBER
private static final String ARG_PARAM1 = "param1";
private static final String ARG_PARAM2 = "param2";

// TODO: Rename and change types of parameters
private String mParam1;
private String mParam2;

public FragmentSimpleInterest() {
    // Required empty public constructor
}

// TODO: Rename and change types and number of parameters
public static FragmentSimpleInterest newInstance(String param1, String param2) {
    FragmentSimpleInterest fragment = new FragmentSimpleInterest();
    Bundle args = new Bundle();
    args.putString(ARG_PARAM1, param1);
    args.putString(ARG_PARAM2, param2);
    fragment.setArguments(args);
    return fragment;
}

@Override
public void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    if (getArguments() != null) {
        mParam1 = getArguments().getString(ARG_PARAM1);
        mParam2 = getArguments().getString(ARG_PARAM2);
    }
}
```

@Override

```
public View onCreateView(LayoutInflater inflater, ViewGroup container, Bundle savedInstanceState)
{
    View view = inflater.inflate(R.layout.fragment_simple_interest, container, false);

    etP = view.findViewById(R.id.etP);
    etR = view.findViewById(R.id.etR);
    etT = view.findViewById(R.id.etT);
    btnCalc = view.findViewById(R.id.btnCalc);
    tvResult = view.findViewById(R.id.tvResult);
    btnCalc.setOnClickListener(v -> {
        double p = Double.parseDouble(etP.getText().toString());
        double r = Double.parseDouble(etR.getText().toString());
        double t = Double.parseDouble(etT.getText().toString());
        double si = (p * r * t) / 100;
        tvResult.setText("The Simple Interest is: " + si);
    });
    return view;
}
}
```

Fragment_Triangle.java

```
package com.bhupendra.trianglesimpleinterest;

import android.os.Bundle;
import androidx.fragment.app.Fragment;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;

public class FragmentTriangle extends Fragment {
```

```
EditText etBase, etHeight;

Button btnCalc;

TextView tvResult;

// TODO: Rename parameter arguments, choose names that match
// the fragment initialization parameters, e.g. ARG_ITEM_NUMBER
private static final String ARG_PARAM1 = "param1";
private static final String ARG_PARAM2 = "param2";

// TODO: Rename and change types of parameters
private String mParam1;
private String mParam2;

public FragmentTriangle() {
    // Required empty public constructor
}

public static FragmentTriangle newInstance(String param1, String param2) {
    FragmentTriangle fragment = new FragmentTriangle();
    Bundle args = new Bundle();
    args.putString(ARG_PARAM1, param1);
    args.putString(ARG_PARAM2, param2);
    fragment.setArguments(args);
    return fragment;
}

@Override
public void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    if (getArguments() != null) {
        mParam1 = getArguments().getString(ARG_PARAM1);
        mParam2 = getArguments().getString(ARG_PARAM2);
    }
}

@Override
```

```
public View onCreateView(LayoutInflater inflater, ViewGroup container, Bundle savedInstanceState) {  
    View view = inflater.inflate(R.layout.fragment_triangle, container, false);  
    etBase = view.findViewById(R.id.etBase);  
    etHeight = view.findViewById(R.id.etHeight);  
    btnCalc = view.findViewById(R.id.btnCalc);  
    tvResult = view.findViewById(R.id.tvResult);  
    btnCalc.setOnClickListener(v -> {  
        double b = Double.parseDouble(etBase.getText().toString());  
        double h = Double.parseDouble(etHeight.getText().toString());  
        double area = 0.5 * b * h;  
        tvResult.setText("The Area of Triangle is: " + area);  
    });  
    return view;  
}  
}
```

MainActivity.java

```
package com.bhupendra.trianglesimpleinterest;  
  
import androidx.appcompat.app.AppCompatActivity;  
import androidx.fragment.app.Fragment;  
import android.os.Bundle;  
import android.view.Menu;  
import android.view.MenuItem;  
  
public class MainActivity extends AppCompatActivity {  
    @Override  
    protected void onCreate(Bundle savedInstanceState) {  
        super.onCreate(savedInstanceState);  
        setContentView(R.layout.activity_main);  
    }  
    @Override  
    public boolean onCreateOptionsMenu(Menu menu) {
```



```

        getMenuInflater().inflate(R.menu.menu_main, menu);

        return true;
    }

    @Override
    public boolean onOptionsItemSelected(MenuItem item) {

        int id = item.getItemId();

        Fragment fragment = null;

        if (id == R.id.menu_si) {

            fragment = new FragmentSimpleInterest();
        } else if (id == R.id.menu_area) {

            fragment = new FragmentTriangle();
        }

        if (fragment != null) {

            getSupportFragmentManager().beginTransaction()

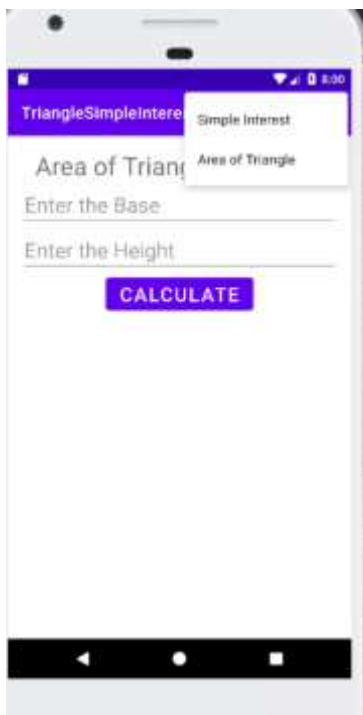
                .replace(R.id.fragment_container, fragment)

                .commit();

        }

        return true;
    }
}

```



b. Demonstrate context menu and popup menu.**Layout.xml**

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"

    android:layout_width="match_parent"

    android:layout_height="match_parent"

    android:orientation="vertical">

    <TextView

        android:layout_width="match_parent"

        android:layout_height="wrap_content"

        android:text="Menus are Here"

        android:textSize="35sp"

        android:layout_margin="15dp"

        android:textAlignment="center"/>

    <Button

        android:layout_width="wrap_content"

        android:layout_height="wrap_content"

        android:text="PopUp Menu"

        android:id="@+id/btn"

        android:textSize="25sp"

        android:layout_margin="10dp"

        android:layout_gravity="center"/>

</LinearLayout>
```

Popup.xml

```
<?xml version="1.0" encoding="utf-8"?>

<menu xmlns:android="http://schemas.android.com/apk/res/android">

    <item

        android:title="Copy"

        android:id="@+id/copy"/>

</menu>
```

```
<item
    android:title="Paste"
    android:id="@+id/paste"/>
<item
    android:title="Cut"
    android:id="@+id/cut"/>
</menu>
```

MainActivity.java

```
package com.bhupendra.contextpopupmenu;

import android.os.Bundle;
import android.view.ContextMenu;
import android.view.Menu;
import android.view.MenuInflater;
import android.view.MenuItem;
import android.view.View;
import android.widget.Button;
import android.widget.PopupMenu;
import android.widget.Toast;
import androidx.activity.EdgeToEdge;
import androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity implements PopupMenu.OnMenuItemClickListener {

    Button btn;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
```

```
        setContentView(R.layout.layout);

        btn=findViewById(R.id.btn);

        registerForContextMenu(btn);

        btn.setOnClickListener(new View.OnClickListener() {

            @Override

            public void onClick(View v) {

                PopupMenu popup=new PopupMenu(getApplicationContext(),v);

                popup.inflate(R.menu.popup);

                popup.setOnMenuItemClickListener(MainActivity.this);

                popup.show();

            }

        });

    }

    public boolean onOptionsItemSelected(MenuItem item){

        if (item.getItemId()==R.id.copy){

            Toast.makeText(this, "Copy is Clicked", Toast.LENGTH_SHORT).show();

        }

        else if (item.getItemId()==R.id.paste){

            Toast.makeText(this, "Paste is Clicked", Toast.LENGTH_SHORT).show();

        }

        else if (item.getItemId()==R.id.cut){

            Toast.makeText(this, "Cut is Clicked", Toast.LENGTH_SHORT).show();

        }

        return true;

    }

    @Override

    public boolean onCreateOptionsMenu(Menu menu) {

        MenuInflater mf = new MenuInflater(this);

        mf.inflate(R.menu.popup, menu);

        return super.onCreateOptionsMenu(menu);

    }

}
```

```
}

@Override

public boolean onOptionsItemSelected(@NonNull MenuItem item) {

    if (item.getItemId()==R.id.copy){

        Toast.makeText(this, "Copy is Clicked!", Toast.LENGTH_SHORT).show();

    }

    else if (item.getItemId()==R.id.paste){

        Toast.makeText(this, "Paste is Clicked!", Toast.LENGTH_SHORT).show();

    }

    else if (item.getItemId()==R.id.cut){

        Toast.makeText(this, "Cut is Clicked!", Toast.LENGTH_SHORT).show(); }

    return super.onOptionsItemSelected(item); }

@Override

public void onCreateContextMenu(ContextMenu menu, View v, ContextMenu.ContextMenuInfo
menuInfo) {

    super.onCreateContextMenu(menu, v, menuInfo);

    MenuInflater mf = new MenuInflater(this);

    mf.inflate(R.menu.popup, menu);}

@Override

public boolean onOptionsItemSelected(@NonNull MenuItem item) {

    if (item.getItemId()==R.id.copy){

        Toast.makeText(this, "Copy is Clicked", Toast.LENGTH_SHORT).show(); }

    else if (item.getItemId()==R.id.paste){

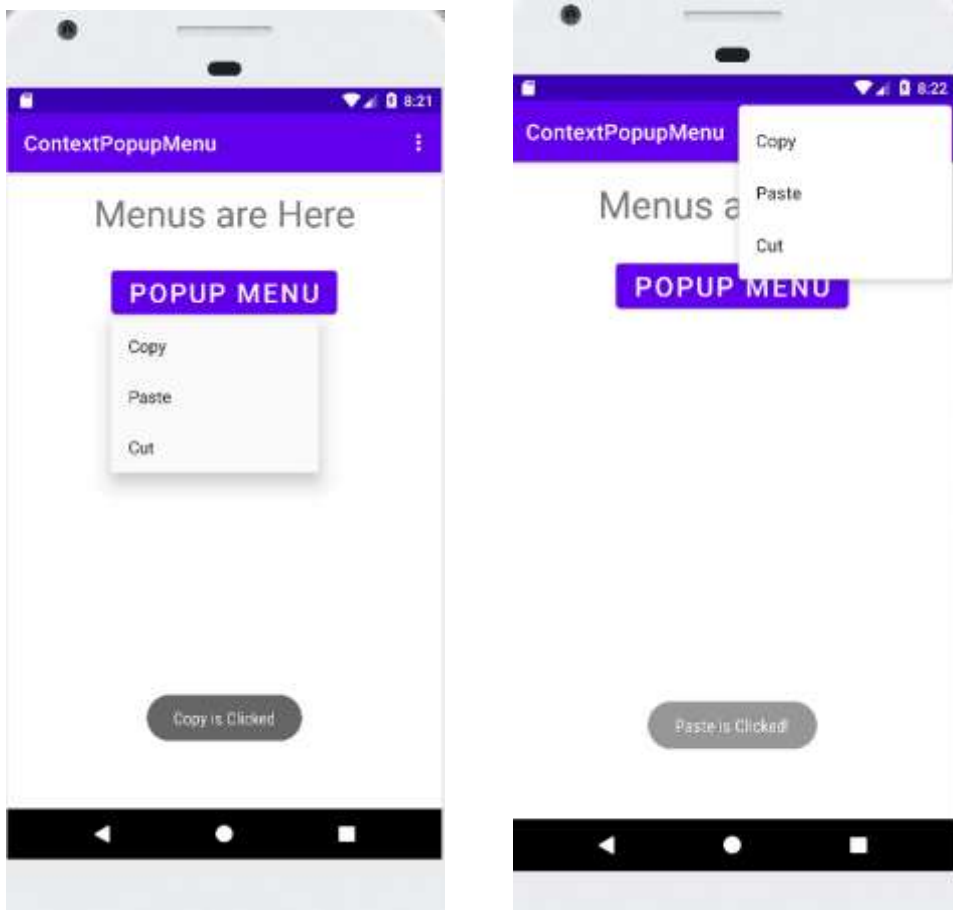
        Toast.makeText(this, "Paste is Clicked", Toast.LENGTH_SHORT).show(); }

    else if (item.getItemId()==R.id.cut){

        Toast.makeText(this, "Cut is Clicked", Toast.LENGTH_SHORT).show(); }

    return super.onOptionsItemSelected(item);

}}
```



c. Create an android app with custom dialog box. Dialog box should ask for Student details (Name,roll,MP,NP,AJ,AE,DS).When save button is pressed. Student details should be displayed in activity along with result and grade.

Activity.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:padding="24dp"
    android:layout_width="match_parent"
    android:layout_height="match_parent">
    <Button
        android:id="@+id/btnShowDialog"
        android:text="Student Details"
        android:layout_width="wrap_content"
```

```
        android:textSize="25sp"

        android:layout_gravity="center"

        android:layout_height="wrap_content"/>

<TextView

        android:id="@+id/tvStudentDetails"

        android:layout_marginTop="20dp"

        android:textSize="20sp"

        android:textColor="#000000"

        android:layout_width="match_parent"

        android:layout_height="wrap_content"/>

</LinearLayout>
```

Customdialog_student.xml

```
<?xml version="1.0" encoding="utf-8"?>

<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"

        android:layout_width="match_parent"

        android:layout_height="wrap_content">

    <LinearLayout

        android:layout_width="match_parent"

        android:layout_height="wrap_content"

        android:orientation="vertical"

        android:padding="20dp">

        <EditText

            android:layout_width="match_parent"

            android:layout_height="wrap_content"

            android:id="@+id/etName"

            android:textSize="25dp"

            android:hint="Name"/>

        <EditText

            android:layout_width="match_parent"

            android:layout_height="wrap_content"
```

```
        android:id="@+id/etRoll"
        android:textSize="25dp"
        android:hint="Roll Number"/>
<EditText
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:id="@+id/etMP"
    android:textSize="25dp"
    android:hint="Mobile Programming Marks"
    android:inputType="number"/>
<EditText
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:id="@+id/etNP"
    android:textSize="25dp"
    android:hint="Network Programming Marks"
    android:inputType="number"/>
<EditText
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:id="@+id/etAJ"
    android:textSize="25dp"
    android:hint="Advance Java Marks"
    android:inputType="number"/>
<EditText
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:id="@+id/etAE"
    android:textSize="25dp"
    android:hint="Applied Economics Marks"
```



```
        android:inputType="number"/>
    <EditText
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:id="@+id/etDS"
        android:textSize="25dp"
        android:hint="Distributed System Marks"
        android:inputType="number"/>
</LinearLayout>
</ScrollView>
```

MainActivity.java

```
package com.bhupendra.studentcustomdialog;

import androidx.appcompat.app.AppCompatActivity;
import androidx.appcompat.app.AlertDialog;
import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.widget.*;

public class MainActivity extends AppCompatActivity {

    Button btnShowDialog;
    TextView tvStudentDetails;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity);

        btnShowDialog = findViewById(R.id.btnShowDialog);
        tvStudentDetails = findViewById(R.id.tvStudentDetails);
        btnShowDialog.setOnClickListener(v -> showStudentDialog());
    }

    private void showStudentDialog() {
```

```
View view = LayoutInflater.from(this).inflate(R.layout.customdialog_student, null);

EditText etName = view.findViewById(R.id.etName);

EditText etRoll = view.findViewById(R.id.etRoll);

EditText etMP = view.findViewById(R.id.etMP);

EditText etNP = view.findViewById(R.id.etNP);

EditText etAJ = view.findViewById(R.id.etAJ);

EditText etAE = view.findViewById(R.id.etAE);

EditText etDS = view.findViewById(R.id.etDS);

AlertDialog.Builder builder = new AlertDialog.Builder(this);

builder.setTitle("Enter Student Details");

builder.setView(view);

builder.setPositiveButton("Save", (dialog, which) -> {

    String name = etName.getText().toString();

    String roll = etRoll.getText().toString();

    int mp = Integer.parseInt(etMP.getText().toString());

    int np = Integer.parseInt(etNP.getText().toString());

    int aj = Integer.parseInt(etAJ.getText().toString());

    int ae = Integer.parseInt(etAE.getText().toString());

    int ds = Integer.parseInt(etDS.getText().toString());

    int total = mp + np + aj + ae + ds;

    float avg = total / 5f;

    boolean pass = mp >= 40 && np >= 40 && aj >= 40 && ae >= 40 && ds >= 40;

    String result = pass ? "Pass" : "Fail";

    String grade;

    if (avg >= 90) grade = "A+";

    else if (avg >= 80) grade = "A";

    else if (avg >= 70) grade = "B";

    else if (avg >= 60) grade = "C";

    else grade = "D";

    String output = "Name: " + name + "\nRoll: " + roll + "\nMarks:\nMobile Programming="
```

```

+ mp + "\nNetwork Programming=" + np + "\nAdvance Java=" + aj + "\nApplied Economics="
+ ae + "\nDistributed System=" + ds + "\nResult: " + result + "\nGrade: " + grade;

tvStudentDetails.setText(output);

});

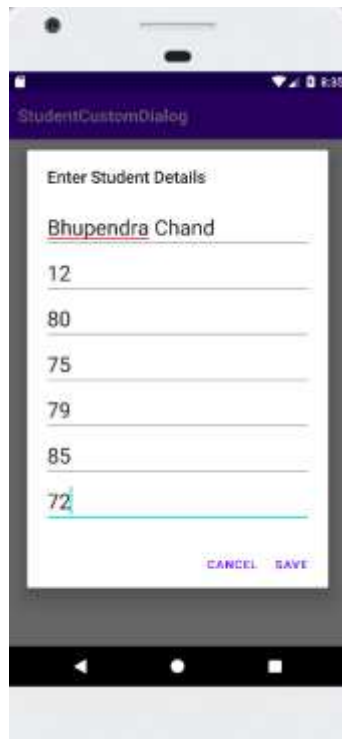
builder.setNegativeButton("Cancel", null);

builder.show();

}

}

```



d. Create a Custom List View with icon.

Layout.xml

```

<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical">

    <ListView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"    android:textSize="30sp"
        android:id="@+id/list"/>

```

```
</LinearLayout>
```

Listitem.xml

```
<?xml version="1.0" encoding="utf-8"?>

<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent" android:layout_height="match_parent">

    <ImageView android:layout_width="100sp"
        android:layout_height="100sp"
        android:contentDescription="TODO"
        android:src="@drawable/pic1" android:layout_margin="10sp"
        android:id="@+id/image"/>

    <TextView

        android:layout_width="match_parent"
        android:layout_height="wrap_content"

        android:text="main title"

        android:textSize="30sp"

        android:layout_margin="8sp"

        android:id="@+id/main" android:layout_toRightOf="@+id/image"/>

    <TextView

        android:layout_width="match_parent"
        android:layout_height="wrap_content"

        android:text="sub title"

        android:textSize="25sp"

        android:layout_margin="50sp"

        android:id="@+id/sub_main"

        android:layout_toRightOf="@+id/image"/>

</RelativeLayout>
```

MainActivity.java

```
import android.os.Bundle;

import android.widget.ListView;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
```

```
    ListView list; @Override

    protected void onCreate(Bundle savedInstanceState) {
super.onCreate(savedInstanceState);

    setContentView(R.layout.layout);

    list = findViewById(R.id.list); // Correct initialization
    String[] main = {"Php", "JavaScript", "React"};

    int[] image = {R.drawable.pic1, R.drawable.pic2, R.drawable.pic3};
    mylistadapter adapter = new mylistadapter(this, main, image);

    list.setAdapter(adapter);

    }
}
```

MyListAdapter.java

```
import android.app.Activity;

import android.content.Context; import
    android.view.LayoutInflater; import
    android.view.View;

import android.view.ViewGroup; import
    android.widget.ArrayAdapter; import
    android.widget.ImageView; import
    android.widget.TextView;

public class mylistadapter extends ArrayAdapter<String> {

    Activity context;

    String[] title; String[] sub_main; int[]
        image;

    public mylistadapter(Activity context, String[] title, int[] image) {

super(context, R.layout.listitem, title);

        this.context = context;

        this.title = title;
```

```
this.image = image;

}

public View getView(int position, View view, ViewGroup parent) {

    LayoutInflater inflater = context.getLayoutInflater();

    View rowView = inflater.inflate(R.layout.listitem, null, true);

    TextView tmain = (TextView) rowView.findViewById(R.id.main);

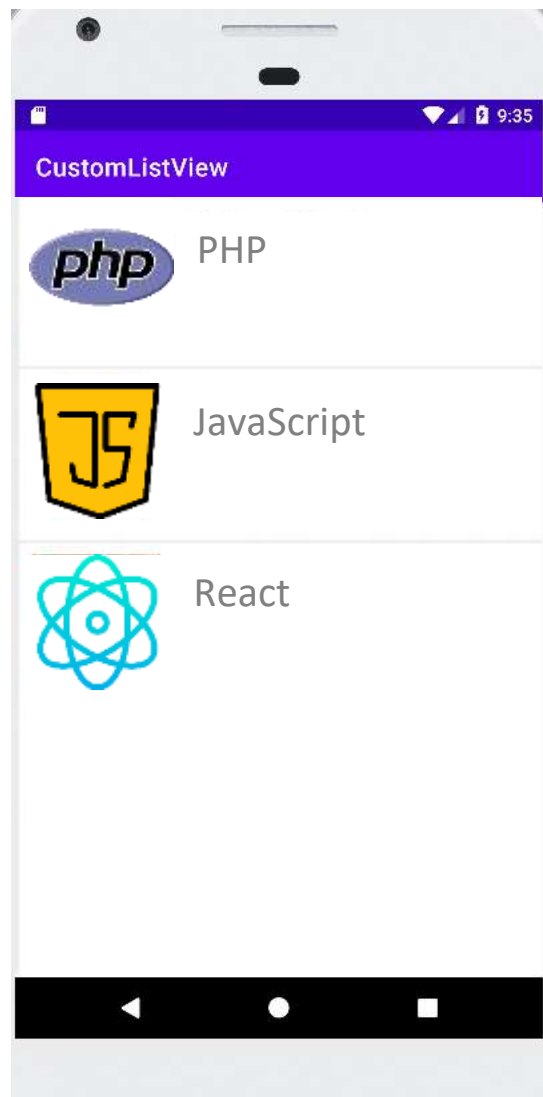
    ImageView timage = (ImageView) rowView.findViewById(R.id.image); tmain.setText(title[position]);

    timage.setImageResource(image[position]);

    tsub_main.setText(sub_main[position]); return rowView;

}

}
```



e. Create a custom Grid View with icon.

Grid_item.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:orientation="vertical"
    android:padding="10dp">
    <ImageView
        android:id="@+id/image"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:src="@drawable/ic_launcher_background"
        android:layout_gravity="center"/>
    <TextView
        android:id="@+id/text"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_gravity="center"
        android:textSize="30sp"
        android:textColor="#000000"
        android:gravity="center"/>
</LinearLayout>
```

ActivityMain.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">
    <GridView
        android:id="@+id/gridView"
        android:layout_width="match_parent"
        android:layout_height="match_parent"
```

```
android:numColumns="auto_fit"
android:columnWidth="100dp"
android:horizontalSpacing="10dp"
android:verticalSpacing="10dp"
android:stretchMode="columnWidth"/>
</RelativeLayout>
```

MyGridAdapter.java

```
import android.app.Activity;
import android.view.LayoutInflater;

import android.view.View;
import android.view.ViewGroup;
import android.widget.ArrayAdapter;
import android.widget.ImageView; import
android.widget.TextView;
import androidx.annotation.NonNull;

class MyGridAdapter extends ArrayAdapter<String> {

    private final Activity context; private final String[] text;
    private final int[] image;
public MyGridAdapter(Activity context, String[] text, int[] image)
{ super(context, R.layout.grid_item, text);

    this.context = context;

    this.text = text;

    this.image = image;
}

@NonNull @Override

    public View getView(int position, @NonNull View convertView, @NonNull ViewGroup parent) {
ViewHolder holder;
    if (convertView == null) {

        LayoutInflater inflater = context.getLayoutInflater(); convertView =
```

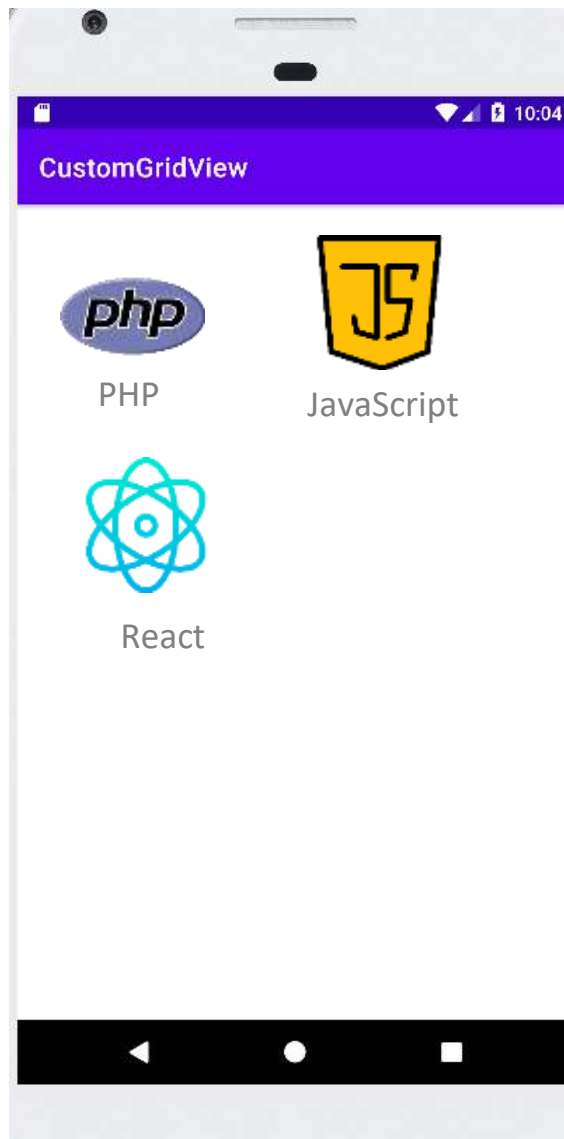


```
inflater.inflate(R.layout.grid_item, parent, false);

holder = new ViewHolder();
holder.imageView = convertView.findViewById(R.id.image);
holder.textView = convertView.findViewById(R.id.text);
convertView.setTag(holder);
} else {
holder = (ViewHolder) convertView.getTag();
}

holder.imageView.setImageResource(image[position]);
holder.textView.setText(text[position]);
return convertView;
}

static class ViewHolder {
    ImageView imageView;
    TextView textView; }
```



f. Create a RecyclerView with icon and event

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical">

    <androidx.recyclerview.widget.RecyclerView
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:scrollbars="vertical" android:id="@+id/list"/>

</LinearLayout>
```

list_item.xml

```
<?xml version="1.0" encoding="utf-8"?>

<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="wrap_content">

    <ImageView
        android:layout_width="100sp"
        android:layout_height="100sp"
        android:layout_alignParentTop="true"
        android:layout_margin="10sp"
        android:id="@+id/image"/>

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:textSize="30sp"
        android:layout_alignParentTop="true"
        android:layout_toRightOf="@+id/image"
        android:id="@+id/txtName" />
```

```

<TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:textSize="25sp"
    android:layout_below="@id/txtName"
    android:layout_toRightOf="@+id/image"
    android:id="@+id/txtAddress" />
</RelativeLayout>

```

MainActivity.java

```

package com.example.recyclerview;

import androidx.appcompat.app.AppCompatActivity;
import androidx.recyclerview.widget.LinearLayoutManager;
import androidx.recyclerview.widget.RecyclerView;
import android.os.Bundle;

public class MainActivity extends AppCompatActivity {
    RecyclerViewAdapter adapter;
    RecyclerView.LayoutManager layoutManager;
    RecyclerView list;
    @Override
    protected void onCreate(Bundle savedInstanceState)
    { super.onCreate(savedInstanceState);
    setContentView(R.layout.layout); list =findViewById(R.id.list);
    String[] name={"Php","JavaScript","React"};
    int[] image = {R.drawable.img1,R.drawable.img2,R.drawable.img3};
    layoutManager = new LinearLayoutManager(MainActivity.this);
    list.setLayoutManager(layoutManager);
    adapter = new RecyclerViewAdapter(MainActivity.this,name,address,image);
    list.setAdapter(adapter);
    }
}

```

```
RecyclerViewAdapter.java package
com.example.recyclerview; import
android.app.Activity; import
android.view.LayoutInflater; import
android.view.View; import
android.view.ViewGroup; import
android.widget.ImageView; import
android.widget.TextView; import
androidx.annotation.NonNull; import
androidx.recyclerview.widget.RecyclerView; public
class RecyclerViewAdapter extends
RecyclerView.Adapter<RecyclerViewAdapter.ViewHolder>{
Activity context;
String[] name;
image;
public RecyclerViewAdapter(Activity context,String[] name,String[] address, int[]
image){ this.context =context; this.name = name;
this.image = image;
}
@NonNull
@Override
public ViewHolder onCreateViewHolder(@NonNull ViewGroup parent, int viewType) {
LayoutInflater inflater = LayoutInflater.from(context);
View listItem = inflater.inflate(R.layout.list_item,parent,false);
ViewHolder viewHolder = new ViewHolder(listItem); return
viewHolder;
}
@Override
public void onBindViewHolder(@NonNull ViewHolder holder, int position)
{ holder.txtName.setText(name[position]);
```

```
holder.txtAddress.setText(address[position]);  
holder.imageView.setImageResource(image[position]);  
}  
  
@Override public int  
getItemCount() { return  
name.length;  
}  
  
public static class ViewHolder extends RecyclerView.ViewHolder{  
    TextView txtName;  
    ImageView  
    ViewHolder(View  
    listItem){ super(listItem);  
    txtName = listItem.findViewById(R.id.txtName);  
    imageView = listItem.findViewById(R.id.image);  
    }  
}
```



